PG No. 45 $C_{6h}(c)$ $6/m$ [hexagonal]

Table 1: Representation matrices

Irrep.						
A_g	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$6^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$6^+_{001}:\begin{bmatrix}1\end{bmatrix}$
	$-1:\begin{bmatrix}1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}1\end{bmatrix}$	$-6^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$-6^+_{001}:\begin{bmatrix}1\end{bmatrix}$
B_g	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}-1\end{bmatrix}$	$6^-_{001}:\begin{bmatrix}-1\end{bmatrix}$	$6^+_{001}:\begin{bmatrix}-1\end{bmatrix}$
	$-1:\begin{bmatrix}1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-6^-_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-6^+_{001}:\begin{bmatrix}-1\end{bmatrix}$
$E^{(a)}_{1g}$	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}-\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}-\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$2_{001}:\begin{bmatrix}-1\end{bmatrix}$	$6^-_{001}:\begin{bmatrix}\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$6^+_{001}:\begin{bmatrix}\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$
	$-1:\begin{bmatrix}1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}-\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}-\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-6^-_{001}:\begin{bmatrix}\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$-6^+_{001}:\begin{bmatrix}\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$
$E^{(b)}_{1g}$	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}-\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}-\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$2_{001}:\begin{bmatrix}-1\end{bmatrix}$	$6^-_{001}:\begin{bmatrix}\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$6^+_{001}:\begin{bmatrix}\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$
	$-1:\begin{bmatrix}1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}-\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}-\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-6^-_{001}:\begin{bmatrix}\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$-6^+_{001}:\begin{bmatrix}\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$
$E^{(a)}_{2g}$	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}-\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}-\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$6^-_{001}:\begin{bmatrix}-\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$6^+_{001}:\begin{bmatrix}-\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$
	$-1:\begin{bmatrix}1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}-\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}-\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}1\end{bmatrix}$	$-6^-_{001}:\begin{bmatrix}-\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$-6^+_{001}:\begin{bmatrix}-\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$
$E^{(b)}_{2g}$	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}-\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}-\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$6^-_{001}:\begin{bmatrix}-\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$6^+_{001}:\begin{bmatrix}-\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$
	$-1:\begin{bmatrix}1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}-\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}-\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}1\end{bmatrix}$	$-6^-_{001}:\begin{bmatrix}-\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$-6^+_{001}:\begin{bmatrix}-\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$
A_u	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$6^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$6^+_{001}:\begin{bmatrix}1\end{bmatrix}$
	$-1:\begin{bmatrix}-1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-6^-_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-6^+_{001}:\begin{bmatrix}-1\end{bmatrix}$
B_u	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}-1\end{bmatrix}$	$6^-_{001}:\begin{bmatrix}-1\end{bmatrix}$	$6^+_{001}:\begin{bmatrix}-1\end{bmatrix}$
	$-1:\begin{bmatrix}-1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}1\end{bmatrix}$	$-6^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$-6^+_{001}:\begin{bmatrix}1\end{bmatrix}$
$E^{(a)}_{1u}$	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}-\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}-\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$2_{001}:\begin{bmatrix}-1\end{bmatrix}$	$6^-_{001}:\begin{bmatrix}\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$6^+_{001}:\begin{bmatrix}\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$
	$-1:\begin{bmatrix}-1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}1\end{bmatrix}$	$-6^-_{001}:\begin{bmatrix}-\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$-6^+_{001}:\begin{bmatrix}-\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$
$E^{(b)}_{1u}$	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}-\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}-\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$2_{001}:\begin{bmatrix}-1\end{bmatrix}$	$6^-_{001}:\begin{bmatrix}\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$6^+_{001}:\begin{bmatrix}\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$
	$-1:\begin{bmatrix}-1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}1\end{bmatrix}$	$-6^-_{001}:\begin{bmatrix}-\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$-6^+_{001}:\begin{bmatrix}-\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$
$E^{(a)}_{2u}$	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}-\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}-\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$6^-_{001}:\begin{bmatrix}-\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$6^+_{001}:\begin{bmatrix}-\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$
	$-1:\begin{bmatrix}-1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-6^-_{001}:\begin{bmatrix}\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$-6^+_{001}:\begin{bmatrix}\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$
$E^{(b)}_{2u}$	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}-\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}-\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$6^-_{001}:\begin{bmatrix}-\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$6^+_{001}:\begin{bmatrix}-\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$
	$-1:\begin{bmatrix}-1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-6^-_{001}:\begin{bmatrix}\frac{1}{2}-\frac{\sqrt{3}i}{2}\end{bmatrix}$	$-6^+_{001}:\begin{bmatrix}\frac{1}{2}+\frac{\sqrt{3}i}{2}\end{bmatrix}$